

Curriculum Vitae

Christopher Ryan King, MD, PhD
Washington University, Department of Anesthesiology
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CV Last updated: 14 Apr 2026

Executive summary

Clinical-educational contributions

I serve in the General Anesthesiology Division at Barnes-Jewish Hospital and Progress West Hospital. Clinical service comprises 85% of my time. I am a physician leader (Pod Leader) for the 19-OR area for Spine, Neurosurgery, Otolaryngology, Hepatobiliary, and Transplant Surgery services at BJH. My sub-specialty expertise focuses on abdominal transplant and vascular anesthesia, providing care for the most liver transplants at BJH of any single anesthesiologist (21% of the total in 2025). Other contributions to the liver group include creating and maintaining the resident protocol and educational materials for transplantation, modifying the anesthetic protocol for total robotic liver transplantation (the first reported in the world), and the donor and recipient protocols for living adult donors. I supervise and personally perform in all inpatient adult ORs and procedure areas at BJH other than cardiac surgery and obstetrics. I supervise and personally perform at Progress West Hospital, including regional anesthesia, obstetrics, and pediatrics. I supervise anesthesiology residents and SRNAs on 67% of clinical days. I am a regular moderator for resident-lead morbidity and mortality presentations and lectures.

Research contributions

Although I have no current NIH awards as PI, I created a significant research program focused on leveraging machine learning (ML) and information technology to improve anesthesia care.

Leadership in Large-Scale Clinical Trials: I was first author for two large RCTs of perioperative telemedicine (ACTFAST-3, TECTONICS; 98,000 patients). These trials were pivotal in building departmental competency in EHR-based trials, implementing real-time ML pipelines, advanced ML model development, and qualitative user feedback analysis.

Advancing ML in Clinical Practice: I led projects to develop and deploy ML models for predicting postoperative complications, guiding clinical decision-making, and improving patient handoffs. This includes developing novel neural network architectures, creating integrated patient databases, and leading multidisciplinary teams to address challenges such as data quality, model robustness, and algorithmic bias. I currently lead a project applying AI to improve peripartum care, including developing a data extraction engine and ML models to enhance electronic fetal monitoring (EFM) accuracy and interpretability. I am a co-investigator on the AHRQ-funded ECHO project, incorporating ML into OR-ICU handoffs to improve communication and patient safety. I submitted projects with others in the department using advanced ML for precision-medicine planning of perioperative care to mitigate severe postoperative pain and opioid consumption.

Research Leadership & Collaboration: I serve as the quantitative methods core leader for the Center for Perioperative Mental Health (P50) renewal, including designing the statistical plan and power analysis for 3 R34 projects and the R01 signature project. I serve as the Division of Clinical and Translational Research Regulatory Scientific Review Chair, leading pre-IRB scientific project review. I have mentored multiple graduate students and post-doctoral scholars, with mentees successfully securing positions in academia and industry. In academic 2024, two students defended their PhD dissertations.

Citizenship: United States of America

Present Position: Associate Professor, Washington University in St Louis School of Medicine

Education

2020-2021 BJC Innovation Lab Fellow
2014-2019 Anesthesiology Residency and Abdominal Organ Transplant Fellowship,
Barnes-Jewish Hospital, Washington University Consortium
2006-2014 MD, University of Chicago Pritzker School of Medicine.
2008-2012 PhD, University of Chicago Department of Health Studies.
Advisor: Dan Nicolae
Committee Chair: Paul Rathouz
2001-2005 BS, University of Virginia. Physics and Mathematics.

Positions and Employment

2005 - 2006 Research assistant, University of Virginia, Department of Psychiatry, Charlottesville, VA
2006 - 2014 MD/PhD student, University of Chicago, Medical Scientist Training Program, Chicago, IL
2007 - 2011 Teaching Assistant (epidemiologic methods, medical epidemiology, principles of
epidemiology, principles of physiology, summer program in outcomes research),
University of Chicago, Pritzker School of Medicine
2014 - 2018 Anesthesiology Resident, Washington University, Department of Anesthesiology, St
Louis, MO
2018 - 2019 Abdominal Organ Transplant Fellow, Washington University, Department of
Anesthesiology, St Louis, MO
2019 - 2020 Instructor, Washington University, Department of Anesthesiology
2021 - 2026 Assistant Professor, Washington University, Department of Anesthesiology
2026 - p Associate Professor, Washington University, Department of Anesthesiology

University and Hospital Appointments and Privileges:

2019, Clinical faculty, Anesthesiology, Barnes-Jewish Hospital, St Louis, MO. Attending Physician.
Privileges for general anesthesiology, liver transplantation, TEE.

2025, Clinical faculty, Anesthesiology, Progress West Hospital. O'Fallon, MO. Attending Physician.
Privileges for general anesthesiology.

Medical Licensure and Board Certification

2016 Permanent Medical License, Missouri State Board of Registration For the Healing Arts
2018 Examination of special competence in basic perioperative transesophageal
echocardiography, National Board of Echocardiography
2019 Diplomat, American Board of Anesthesiology

Editorial Responsibilities

2018 Reviewer, British Journal of Anaesthesia
2020 Reviewer, Journal of Clinical Anaesthesia
2022 Reviewer, Anesthesiology
2022 Reviewer, The Joint Commission Journal on Quality and Patient Safety
2022 Reviewer, BMJ Open
2023 Reviewer, Journal of biomedical informatics

2024 Reviewer, PLoS One
2024 Reviewer, JMIR Aging
2025 Reviewer, JMIR Medical Informatics

Other Peer Review Activities

10/2023 Reviewer. Wash U Clinical and Translational Research Funding Program grants
4/2025, 4/2026 Reviewer. Wash U AI for Health seed grants

Professional Societies And Organizations:

2012 - 2014 Member, American Statistical Association
2014 - 2022 Member, American Society of Anesthesiologists
2016 - 2023 Member, International Anesthesia Research Society
2017 - 2026 Member, International Liver Transplantation Society

Research Support:

ACTIVE:

- a. Governmental.
 - i. 1R01HS029324-01 (PI: Abraham) Role: Co-I
- b. Non-governmental
 - None

PAST:

- a. Governmental.
 - i. F30 HL103105-01. "Methods for Genetic Resequencing Association Studies: Training and Research." Role: PI.
 - ii. T32 GM007281-38. Clark, Marcus Ramsay (PI) 07/01/75-06/30/13. Medical Scientist National Research Service Award. Role: TA
 - iii. T32 GM108539-05. Evers, Alex (PI) 10/01/17-09/30/19 TRAINING PROGRAM IN ANESTHESIOLOGY RESEARCH Role: Post-Doctoral Scholar
 - iv. 5KL2TR002346-04 (KL2 Scholar)
- b. Non-governmental
 - i. Big Ideas Competition 2019, Washington University Innovations Lab. Abraham, Joanna (PI) 07/01/19-06/30/20 Machine Learning Risk Predictions at OR-ICU handoff. Role: Co-Investigator
 - ii. Clinical and Translational Research Funding Program: CTRFP1608.
 - iii. WUSTL TRIADS seed funding. Role: PI

PATENTS

Kovatchev, B. P. & King, C. R. Accuracy of Continuous Glucose Sensors. (2008). United States 20080314395

TEACHING TITLE AND RESPONSIBILITIES: Supervision for WUSM students, WUSTL-BJH anesthesiology trainees, Goldfarb School of Nursing SRNA. No courses.

TRAINING / MENTEE RECORD:

Current Trainees / Mentees

None

Prior Trainees / Mentees

Hao Liu. PhD Student (2020 - 2025). Shared supervision with Yixin Chen (WUSTL CSE).

Lecheng Kong. PhD Student (2020 - 2025). Shared supervision with Yixin Chen (WUSTL CSE).

Tripathi, Sandhya. Postdoctoral scholar (2020-2024). Shared supervision with Yixin Chen (WUSTL CSE).

Abdelhack, Mohamed. Postdoctoral scholar (2019-2021). Shared supervision with Yixin Chen (WUSTL CSE).

Shambe, Ayanna. Research assistant / medical student (2021).

Shuting Cui. Master's student (2021-2022). Shared supervision with Yixin Chen (WUSTL CSE).

Zhou Chu. Master's student (2021-2022). Shared supervision with Yixin Chen (WUSTL CSE).

Yukun Wang. Master's student (2021-2022). Shared supervision with Yixin Chen (WUSTL CSE).

Publications

King, C. R. & Fritz, B. A. Machine learning models in anaesthesiology: bridging the gap from model training to implementation. *Br J Anaesth* 136, 1723–1726 (2026).

Friedrichsen, K. A., Judge, B. A., Jones, L. A., Rajamanickam, J., Qiu, L., Soda, A. K., Rahmani, F., Benzinger, G. R., **King, C. R.**, Chaney, A. M., Nickels, M. L., Gropler, R. J., Raji, C. A., White, R. L., Perlmutter, J. S., Laforest, R., An, H., Tu, Z., Benzinger, T. L. S. & Brier, M. R. Evaluation of [11C]CS1P1 in Healthy Young and Older Adults. *AJNR Am J Neuroradiol* 47, 175–183 (2026).

Holzer, K. J., Alaverdyan, H., Xu, Z., Frumkin, M. R., Frey, K. A., Gregory, S. H., Rodebaugh, T. L., Lu, C., **King, C. R.**, Head, D., Kannampallil, T. & Haroutounian, S. Protocol for Personalised Prediction of Persistent Postsurgical Pain. *BMJ Open* 16, e107055 (2026).

King, C. R., Fritz, B. A., Gregory, S. H., Budelier, T. P., Ben Abdallah, A., Kronzer, A., Helsten, D. L., Torres, B., McKinnon, S. L., Tripathi, S., Abdelhack, M., Goswami, S., Montes de Oca, A., Mehta, D., Valdez, M. A., Karanikolas, E., Higo, O., Kerby, P., Henrichs, B., Wildes, T. S., Politi, M. C., Abraham, J., Avidan, M. S. & Kannampallil, T. Effect of telemedicine support for intraoperative anaesthesia care on postoperative outcomes: the TECTONICS randomised clinical trial. *British Journal of Anaesthesia* (2025). doi:10.1016/j.bja.2024.11.017

Abraham J, **King CR**, Pedomallu L, Light M, Henrichs B. Effect of standardized EHR-integrated handoff report on intraoperative communication outcomes. *J Am Med Inform Assoc.* 2024 Jul 31:ocae204. doi: 10.1093/jamia/ocae204. Epub ahead of print. PMID: 39081222.

Fritz BA, **King CR**, Abdelhack M, Chen Y, Kronzer A, Abraham J, Tripathi S, Ben Abdallah A, Kannampallil T, Budelier TP, Helsten D, Montes de Oca A, Mehta D, Sontha P, Higo O, Kerby P, Gregory SH, Wildes TS, Avidan MS. Effect of machine learning models on clinician prediction of postoperative complications: the Perioperative ORACLE randomised clinical trial. *Br J Anaesth.* 2024 Sep 10:S0007-0912(24)00468-9. doi: 10.1016/j.bja.2024.08.004. Epub ahead of print. PMID: 39261226.

Abdelhack, M., Tripathi, S., Chen, Y., Avidan, M. S. & **King, C. R.** Social vulnerability and surgery outcomes: a cross-sectional analysis. *BMC Public Health* 24, 1907 (2024).

Vawda, D. O., **King, C.**, Toit, L. du, Dyer, R. A., Masuku, N. J. & Bishop, D. G. Agreement between three noninvasive temperature monitoring devices during spinal anaesthesia for caesarean delivery: a prospective observational study. *J Clin Monit Comput* (2024). doi:10.1007/s10877-024-01154-1

Khan, A. S., Scherer, M., Panni, R., Cullinan, D., Martens, G., Kangarga, I., **King, C. R.**, Benzinger, R., Wellen, J. R., Chapman, W. C. & Doyle, M. B. Total Robotic Liver Transplant: The Final Frontier of Minimally Invasive Surgery. *Am J Transplant* S1600-6135(24)00240–5 (2024). doi:10.1016/j.ajt.2024.03.030

Tripathi, S., Fritz, B. A., Abdelhack, M., Avidan, M. S., Chen, Y. & **King, C. R.** Multi-view representation learning for tabular data integration using inter-feature relationships. *J Biomed Inform* 151, 104602 (2024).

King, C. R., Gregory, S., Fritz, B. A., Budelier, T. P., Ben Abdallah, A., Kronzer, A., Helsten, D. L., Torres, B., McKinnon, S., Goswami, S., Mehta, D., Higo, O., Kerby, P., Henrichs, B., Wildes, T. S., Politi, M. C., Abraham, J., Avidan, M. S., Kannampallil, T., & ACTFAST Study Group. An Intraoperative Telemedicine Program to Improve Perioperative Quality Measures: The ACTFAST-3 Randomized Clinical Trial. *JAMA Netw Open* 6, e2332517 (2023).

Liu, H., Chen, Y., Fritz, B. & **King, C.** TabContrast: A Local-Global Level Method for Tabular Contrastive Learning. *NeurIPS 2023 Second Table Representation Learning Workshop*.

Abdelhack, M., Zhang, J., Tripathi, S., Fritz, B. A., Felsky, D., Avidan, M., Chen, Y. & **King, C. R.** A Modulation Layer to Increase Neural Network Robustness Against Data Quality Issues. *Transactions on Machine Learning Research* (2023)

Liu, H., **King, C.**, Abraham, J., Palmer, C., Kronzer, A., Konzen, L., White, J., Bertrand, J., Drewry, A., Chen, Y. & Fritz, B. 294: PREDICTING ICU LENGTH OF STAY PRIOR TO ICU ADMISSION USING META-LEARNING. *Critical Care Medicine* 51, 132 (2023).

King, C. R., Shambe, A. & Abraham, J. Potential uses of AI for perioperative nursing handoffs: a qualitative study. *JAMIA Open* 6, ooad015 (2023).

Fritz, B. A., Pugazenthi, S., Budelier, T. P., Tellor Pennington, B. R., **King, C. R.**, Avidan, M. S. & Abraham, J. User-Centered Design of a Machine Learning Dashboard for Prediction of Postoperative Complications. *Anesth Analg* (2023). doi:10.1213/ANE.00000000000006577.

Abraham, J., Bartek, B., Meng, A., **Ryan King, C.**, Xue, B., Lu, C. & Avidan, M. S. Integrating machine learning predictions for perioperative risk management: Towards an empirical design of a flexible-standardized risk assessment tool. *J Biomed Inform* 137, 104270 (2023).

Xue, B., Jiao, Y., Kannampallil, T., Fritz, B., **King, C.**, Abraham, J., Avidan, M. & Lu, C. Perioperative Predictions with Interpretable Latent Representation. in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* 4268–4278 (Association for Computing Machinery, 2022). doi:10.1145/3534678.3539190

Li, D., Xue, B., **King, C.**, Fritz, B., Avidan, M., Abraham, J. & Lu, C. Self-explaining Hierarchical Model for Intraoperative Time Series. in 1041–1046 (IEEE Computer Society, 2022).

doi:10.1109/ICDM54844.2022.00128

Fritz, B., **King, C.**, Chen, Y., Kronzer, A., Abraham, J., Ben Abdallah, A., Kannampallil, T., Budelier, T., Montes de Oca, A., McKinnon, S., & others. Protocol for the perioperative outcome risk assessment with computer learning enhancement (Periop ORACLE) randomized study. *F1000Research* 11, 653 (2022).

Fritz, B. A., **King, C. R.**, Mehta, D., Somerville, E., Kronzer, A., Ben Abdallah, A., Wildes, T., Avidan, M. S., Lenze, E. J., Stark, S., & ENGAGES Research Group. Association of a Perioperative Multicomponent Fall Prevention Intervention With Falls and Quality of Life After Elective Inpatient Surgical Procedures. *JAMA Netw Open* 5, e221938 (2022).

Whitlock EL, Gross ER, **King CR**, Avidan MS. Anaesthetic depth and delirium: a challenging balancing act. *Br J Anaesth.* 2021 Nov;127(5):667-671. doi: 10.1016/j.bja.2021.08.003. Epub 2021 Sep 6. PubMed PMID: 34503835.

Fritz, B.A., **King, C.R.**, Mickle, A.M., Wildes, T.S., Budelier, T.P., Oberhaus, J., Park, D., Maybrier, H.R., Ben Abdallah, A., Kronzer, A., et al. (2021). Effect of electroencephalogram-guided anaesthesia administration on 1-yr mortality: follow-up of a randomised clinical trial. *Br J Anaesth* 127, 386–395.

King, C.R., Abraham, J., Fritz, B.A., Cui, Z., Galanter, W., Chen, Y., and Kannampallil, T. (2021). Predicting self-intercepted medication ordering errors using machine learning. *PLOS ONE* 16, e0254358.

Xue, B., Jiao, Y., Kannampallil, T., Fritz, B., **King, C.**, Abraham, J., Avidan, M. & Lu, C. Perioperative Predictions with Interpretable Latent Representation. in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* 4268–4278 (Association for Computing Machinery, 2022). doi:10.1145/3534678.3539190

Xue, B., Li, D., Lu, C., **King, C.R.**, Wildes, T., Avidan, M.S., Kannampallil, T., and Abraham, J. (2021). Use of Machine Learning to Develop and Evaluate Models Using Preoperative and Intraoperative Data to Identify Risks of Postoperative Complications. *JAMA Netw Open* 4, e212240.

Abraham, J., **King, C. R.** & Meng, A. Ascertaining Design Requirements for Postoperative Care Transition Interventions. *Appl Clin Inform* 12, 107–115 (2021). PMID: 33626584

Budelier, T.P., **King, C.R.**, Goswami, S., Bansal, A., Gregory, S.H., Wildes, T.S., Abraham, J., McKinnon, S.L., Cooper, A., Kangrga, I., et al. (2020). Protocol for a proof-of-concept observational study evaluating the potential utility and acceptability of a telemedicine solution for the post-anesthesia care unit. *F1000Res* 9, 1261. PMID: 33214879

Gregory SH, **King CR**, Ben Abdallah A, Kronzer A, Wildes TS. Abnormal preoperative cognitive screening in aged surgical patients: a retrospective cohort analysis. *Br J Anaesth.* 2020 Sep 14; PMID: 32943193

Zhang M, **King CR**, Avidan M, Chen Y. Hierarchical Attention Propagation for Healthcare Representation Learning. *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining.* New York, NY, USA: Association for Computing Machinery; 2020 [cited 2020 Sep 27]. p. 249–256. Available from:

<https://doi.org/10.1145/3394486.3403067>

Maranhao, B., Scott, A. W., Scott, A. R., Maeng, J., Song, Z., Baddigam, R., **King, C. R.**, McCormick, M., Kangrga, I. & Guffey, R. Probability of fit failure with reuse of N95 mask respirators. *Br J Anaesth* (2020). PMID: 32682553

Fritz, B. A., Abdelhack, M., **King, C. R.**, Chen, Y. & Avidan, M. S. Update to ‘Deep-learning model for predicting 30-day postoperative mortality’ (*Br J Anaesth* 2019; 123: 688-95). *Br J Anaesth* 125, e230–e231 (2020). PMID: 32389391

Fritz, Bradley A., **Christopher R. King**, Arbi Ben Abdallah, Nan Lin, Angela M. Mickle, Thaddeus P. Budelier, Jordan Oberhaus, et al. “Preoperative Cognitive Abnormality, Intraoperative Electroencephalogram Suppression, and Postoperative Delirium: A Mediation Analysis.” *Anesthesiology*, February 5, 2020.

King, C.R., Fritz, B.A., Escallier, K., Ju, Y.-E.S., Lin, N., McKinnon, S., Avidan, M.S., and Palanca, B.J. (2020). Association Between Preoperative Obstructive Sleep Apnea and Preoperative Positive Airway Pressure With Postoperative Intensive Care Unit Delirium. *JAMA Netw Open* 3, e203125.

King, C.R., Abraham, J., Kannampallil, T.G., Fritz, B.A., Ben Abdallah, A., Chen, Y., Henrichs, B., Politi, M., Torres, B.A., Mickle, A., et al. (2019). Protocol for the Effectiveness of an Anesthesiology Control Tower System in Improving Perioperative Quality Metrics and Clinical Outcomes: the TECTONICS randomized, pragmatic trial. *F1000Res* 8, 2032.

Cui, Z., Fritz, B.A., **King, C.R.**, Avidan, M.S., and Chen, Y. (2019). A Factored Generalized Additive Model for Clinical Decision Support in the Operating Room. *AMIA Annu Symp Proc* 2019, 343–352.

Fritz, B.A., Cui, Z., Zhang, M., He, Y., Chen, Y., Kronzer, A., Ben Abdallah, A., **King, C.R.**, and Avidan, M.S. (2019). Deep-learning model for predicting 30-day postoperative mortality. *Br J Anaesth* 123, 688–695.

King, C.R., Escallier, K.E., Ju, Y.-E.S., Lin, N., Palanca, B.J., McKinnon, S.L., and Avidan, M.S. (2019). Obstructive sleep apnoea, positive airway pressure treatment and postoperative delirium: protocol for a retrospective observational study. *BMJ Open* 9, e026649.

Maybrier HR, **King CR**, Crawford A, Emmert DA, Mickle AM, Wildes TS, Avidan MS, Palanca BJA (2019). Early Postoperative Actigraphy Poorly Predicts Hypoactive Delirium. *J Clin Sleep Med*; 15(1):79-87. PMID 30621829.

Henn, M.C., Zajarias, A., Quader, N., Sintek, M., Lasala, J.M., Koogler, K., Damiano, M.S., Kachroo, P., Miller, D.C., **King, C.R.**, et al. (2018). Observed to expected 30-day mortality as a benchmark for transcatheter aortic valve replacement. *J. Thorac. Cardiovasc. Surg.* pii: S0022-5223(18)32031-2. PMID 30454980.

Goo Jun, Alisa K Manning, Marcio Almeida, Matthew Zawistowski, Andrew Robert Wood, Tanya M Teslovich, Christian Fuchsberger, Shuang Feng, Pablo Cingolani, Kyle J Gaulton, Thomas Dyer, Thomas W Blackwell, Han Chen, Peter S Chines, Sungkyoung Choi, Claire Churchhouse, Pierre Fontanillas, **C Ryan King**, SungYoung Lee, Stephen Lincoln, Vasily Trubetskoy, Mark DePristo, Tasha Fingerlin, Robert Grossman, Jason Grundstad, Alison Heath, Jayoun Kim, Young Jin Kim,

Jason Laramie, Jaehoon Lee, Heng Li, Xuanyao Liu, Oren Livne, Adam E Locke, Julian Maller, Alexander Mazur, Andrew P Morris, Toni I Pollin, Derek Ragona, David Reich, Manuel A Rivas, Laura J Scott, Xueling Sim, Richard G Tearle, Yik Ying Teo, Amy L Williams, Sebastian Zollner, Joanne E Curran, Juan Peralta, Beena Akolkar, Graeme Bell, Noel P. Burt, Nancy J. Cox, Jose C. Florez, Craig Hanis, Catherine McKeon, Karen L. Mohlke, Mark Seielstad, James G. Wilson, Gil Atzmon, Jennifer E. Below, Josee Dupuis, Dan L. Nicolae, Donna Lehman, Taesung Park, Sungho Won, Robert Sladek, David Altshuler, Mark I. McCarthy, Ravi Duggirala, Michael Boehnke, Tim M Frayling, Goncalo Abecasis, John Blangero (2017). Evaluating the Contribution of Rare Variants to Complex Disease using Pedigrees. *Proc Natl Acad Sci*; 115(2):379-384. PMID 29279374.

King, C.R., and Nicolae, D.L. (2014). GWAS to Sequencing: Divergence in Study Design and Analysis. *Genes* (Basel) 5, 460–476.

King, CR (2012). The rare variant hypothesis: Interpretation and estimation in sequencing association studies. Ph.D. The University of Chicago.
Topics explored roughly cover those of King (2010) and King and Nicolae (2014) as well as a chapter on semiparametric recovery of the bias presented in King and Nicolae (2014).

King, C.R., Rathouz, P.J., and Nicolae, D.L. (2010). An Evolutionary Framework for Association Testing in Resequencing Studies. *PLoS Genet* 6, e1001202.

King, C.R., Knutson, K.L., Rathouz, P.J., Sidney, S., Liu, K., and Lauderdale, D.S. (2008). Short sleep duration and incident coronary artery calcification. *JAMA*. 300, 2859–2866.

King, C.R., Anderson, S.M., Breton, M.D., Clarke, W.L., and Kovatchev, B.P. (2007). Modeling of calibration effectiveness and blood-to-interstitial glucose dynamics as potential confounders of the accuracy of continuous glucose sensors during hyperinsulinemic clamp. *J Diabetes Sci Technol* 1, 317–322.

Kovatchev, B.P., **King, C.R.**, Breton, M., Anderson, S., and Clarke, W. (2006). Clinical assessment and mathematical modeling of the accuracy of continuous glucose sensors (CGS). *Conf Proc IEEE Eng Med Biol Soc* 1, 71–74.

Research Reports

Liu, H., Zhang, M., Dong, Z., Kong, L., Chen, Y., Fritz, B., Tao, D. & **King, C.** Time Associated Meta Learning for Clinical Prediction. Preprint at <https://doi.org/10.48550/arXiv.2303.02570> (2023)

Kong, L., **King, C.**, Fritz, B. & Chen, Y. A Multi-View Joint Learning Framework for Embedding Clinical Codes and Text Using Graph Neural Networks. Preprint at <https://doi.org/10.48550/arXiv.2301.11608> (2023)

Tripathi, S., Fritz, B. A., Avidan, M. S., Chen, Y. & **King, C. R.** Algorithmic Bias in Machine Learning Based Delirium Prediction. arXiv preprint arXiv:2211.04442 (2022).

Tripathi, S., Fritz, B. A., Abdelhack, M., Avidan, M. S., Chen, Y. & **King, C. R.** Deep Learning to Jointly Schema Match, Impute, and Transform Databases. Preprint at <https://doi.org/10.48550/arXiv.2207.03536> (2022)

King CR, Rathouz PJ, and Nicolae DL (2013). Generalizing from sequencing studies. arXiv:1312.7714 [stat].

Teaching

Washington University Presentations & Lectures

Guest lectures (recurring lectures collapsed)

- Mar 2020. Department of Anesthesiology resident education day, “Frailty and Recovery After Surgery”.
- Sep 2020. Department of Anesthesiology Grand Rounds “Incorporating artificial intelligence into the pre- and post-anesthesia workflow”
- Feb 2021. Department of Anesthesiology resident education day, “Individual provider report cards in cardiac surgery”.
- October 2021. Department of Anesthesiology scientific review session, “AI and disparities in clinical decision-making”
- Nov 2023, repeating 2019-2022. Department of Anesthesiology Research Retreat “Update in Perioperative Risk Prediction and Implementation”
- Nov 2022. Department of Anesthesiology resident education day, “Update in Xenotransplantation”.
- Nov 2023. Repeating 2022. Department of Anesthesiology resident education day “Cardiac evaluation for transplantation”.
- Nov 2024, Repeating 2023. CSE 419 guest lecture “Applied perioperative ML”.
- Nov 2024, Repeating 2023. Department of Anesthesiology CA-2 education day “Rotational thromboelastography”.
- Nov 2025, Repeating 2021-2025. Department of Anesthesiology CA-2 education day “Liver transplantation”.
- Dec 2025, Repeating 2020-2025 Department of Anesthesiology CA-1 education day “Fluids and electrolytes”
- May 2025 Department of Anesthesiology “CA3 POCUS OSCE Training: Transthoracic Cardiac and Lung”
- March 25 2026 Department of Anesthesiology "CA1 POCUS Training: Lung, abdominal"
- April 1 2026 Department of Anesthesiology "CA2 POCUS Training: FAST exam"
- May 2025 TRAIDS Data Science Project Updates “Predictive Learning Algorithms for Neonatal - Maternal Outcome Risk Evaluation Using Cardiotocography and EHR data”
- Jan 2026, repeating 2020-2026. Department of Anesthesiology Grand Rounds “Morbidity and Mortality”.
- Feb 5 2026 Department of Anesthesiology "Intern curriculum lecture: transplantation and transplant call"
- Aug 13 2025 Department of Otolaryngology "Perioperative optimization and head and neck surgery"
- October 15 2025 Department of Anesthesiology "Transplant fellowship series: Cirrhotic Coagulopathy and Monitoring"

Other Conference presentations

Jan 2024 CODS-COMAND 2024, Contrastive Learning: Big Data Foundations and Applications. IIIT Bangalore, Bengaluru, India

Other University and Departmental Service

Department of Anesthesiology, Division of Clinical and Translational Research: Chair of Regulatory Scientific Review

Clinical Monitor for SWIPED RCT (PI Palanca)

Clinical Monitor for SWIPED II RCT (PI Palanca)

Data Safety and Monitoring Board member, TELE-TOC RCT (PI Abraham)